

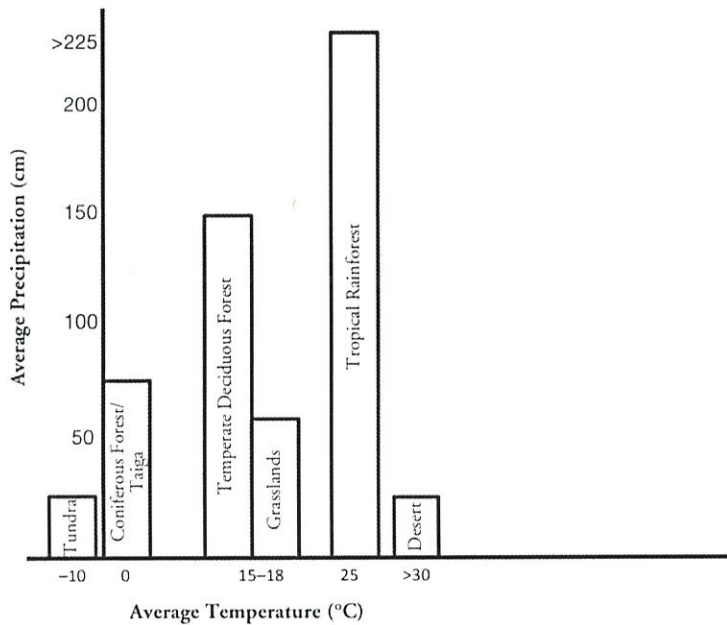
How do temperature and precipitation affect the distribution of plants and animals

Why?

Have you ever wondered why no polar bears are in California, or cacti in Alaska? The amount of precipitation and the temperature varies tremendously across North America, resulting in well-defined community boundaries or **biomes** that are suitable for some populations but not for others.

Model 1 – Biomes of North America

Average Precipitation and Temperature



1. According to Model 1, which two biomes have the same amount of rainfall?

Tundra & Desert

2. Which biome in Model 1 has the most rainfall?

Tropical Rainforest

3. Which biome has the highest average temperature?

Desert

4. Which two biomes have nearly the same average temperature?

Temperate deciduous forest & Grasslands

5. Refer to Model 1.

- Name the three different types of forests shown in Model 1.

1) Coniferous forest / Taiga 2) Temperate deciduous forest 3) Trop. Rainforest

- What is another name for the coniferous forest?

Taiga

6. Refer to Model 1.

- What characteristic differentiates the tundra and the desert? Temperature

- Why might the tundra also be known as the "frozen desert"?

Very Low Precipitation.

7. Consider the biomes in Model 1 with moderate temperatures.

- What characteristic differentiates the grasslands and temperate deciduous forests?

The amount of precipitation

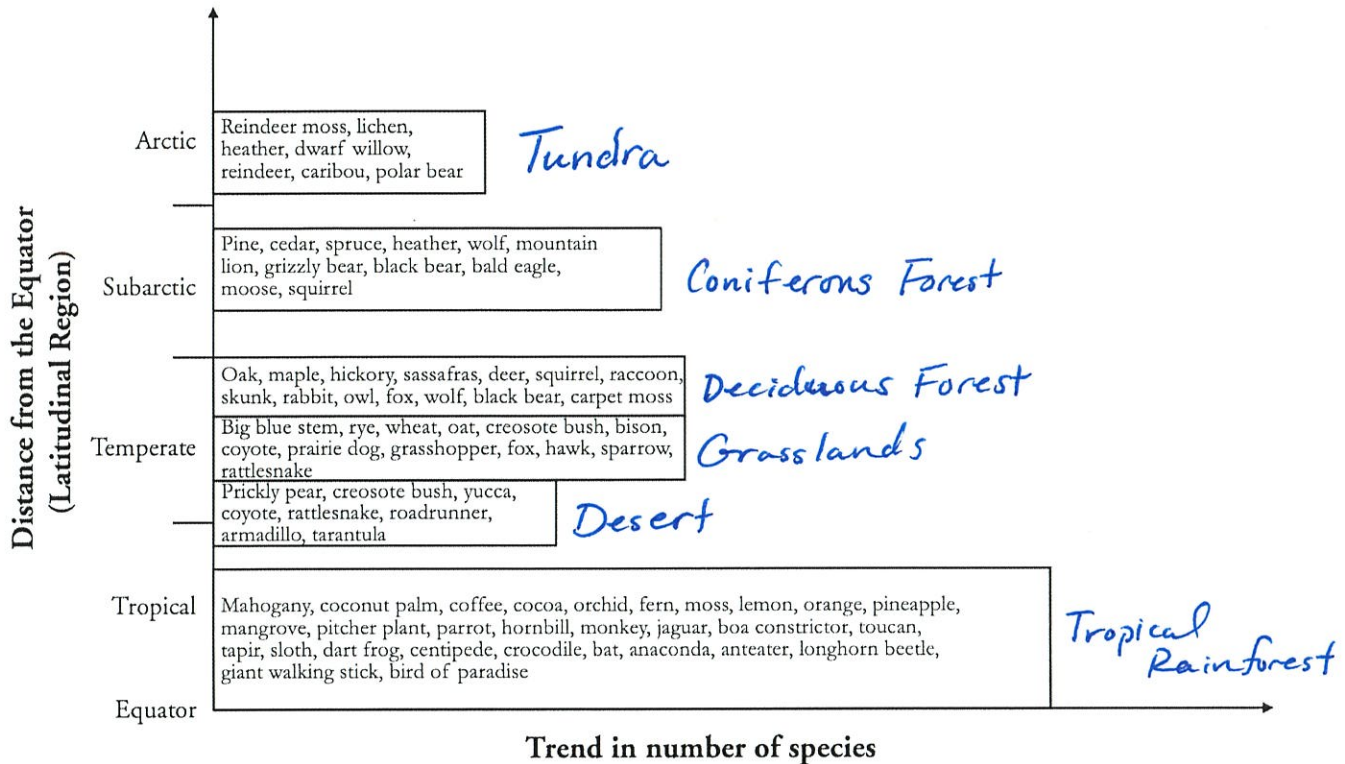
- What would be the most likely reason for the grasslands having grasses rather than trees as their dominant plant species?

Less precipitation. Trees require more water

8. Describe the relationship between temperature, rainfall, and the type of forest.

Cold temperature has Low precipitation = coniferous forest
Temperate (warm) has higher precipitation = deciduous forest
Higher temperature has highest rainfall = tropical rainforest

Model 2 – Plant and Animal Species in North American Biomes



9. The six bars in Model 2 represent the six North American biomes introduced in Model 1, and include a list of some plants and animals typically found in each biome.

- What information is represented by the x-axis in Model 2?

Trend in number of species

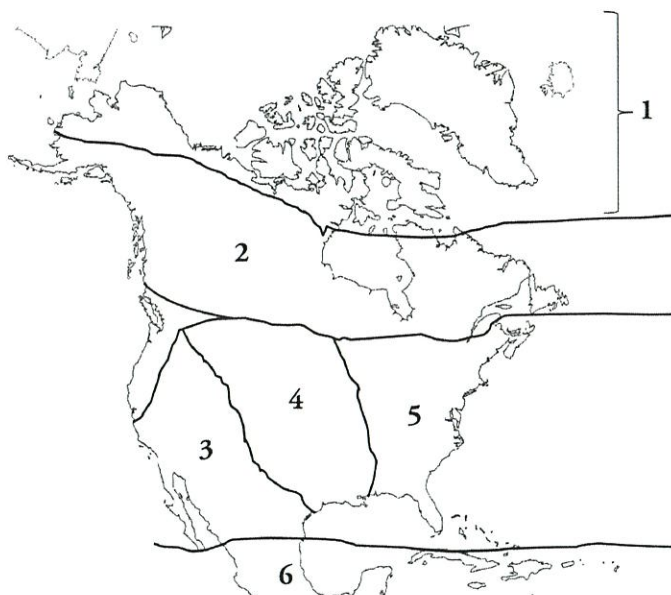
- What information is represented by the y-axis in Model 2?

Distance from equator (Latitude)

- How might the y-axis information in Model 2 be related to one of the data axes in Model 1?

Closer to the equator will likely have higher average temp.

10. Identify each of the six biomes found in North America on the map below. Average rainfall data is given to help you choose between biomes in the same latitudinal region.



Zone	Rainfall (cm)
1	25
2	75
3	25
4	60
5	150
6	>225

1. *Tundra*

2. *Coniferous forest*

3. *Desert*

4. *Grasslands*

5. *Deciduous Forest*

6. *Tropical rainforest*

11. Use information from Model 1 and the map in Question 10 to label the six biomes DIRECTLY in the chart in Model

12. Refer to Model 2.

- Which latitudinal region contains the fewest number of species? *The Arctic region*
- List the biome(s) found in this latitudinal region. *Tundra*
- Which latitudinal region contains the largest number of species? *The tropical region*
- List the biome(s) found in this latitudinal region. *Tropical rainforest & Desert*

13. **Biodiversity** is a term used to describe the variety of plant and animal species in a given biome. An ecosystem or biome that supports a large variety of species is very biodiverse.

- What biome in North America has the highest biodiversity? *Tropical rainforest*
- What characteristics of this biome enable it to support such a high level of biodiversity?
Warmer temps. & Higher amounts of precipitation
- What biome in North America has the lowest biodiversity?
Tundra
- What characteristics of this biome make it difficult for a high level of biodiversity to develop?
Very low temps. & Very little precipitation

14. What is the general trend of the biodiversity in biomes as the latitude moves from the arctic region towards the equator?

Biodiversity generally increases the closer to the equator